

Jordan Cahoon

Personal email: jlcah5@gmail.com

School email: cahoon@stanford.edu

Website: <https://www.jordancahoon.com/>

Education

2024-2029	STANFORD UNIVERSITY PhD in Biomedical Data Science GPA: 4.0
2020-2024	UNIVERSITY OF SOUTHERN CALIFORNIA BS in Computer Science, Minor in Computational Biology & Bioinformatics GPA: 3.86

Research Experience

Jun 2025 - Present	PhD Student. ADVISED BY EMILY ALSENTZER & STEPHEN MONTGOMERY Stanford University, Department of Biomedical Data Science • Building a benchmark dataset to evaluate information retrieval performance in electronic health records, supporting long-context clinical reasoning in large language models. • Developing computational methods that integrate electronic health records and multi-omics data to accelerate rare disease diagnosis.
Feb 2021 - Jun 2024	Undergraduate Researcher. ADVISED BY CHARLESTON CHIANG University of Southern California, Keck School of Medicine • Conducted population genetics research aimed at improving generalizability across ancestrally diverse populations. • Identified discrepancies in genotype imputation accuracy for non-European populations (e.g., East Asian, South Asian, Oceanian, Southeast Asian) using the TOPMed reference panel. • Designed a framework using meta-imputation that improved imputation accuracy for East and Southeast Asian cohorts, with enhanced recovery of population-specific variants. • Developed a deep learning model leveraging genealogical trees to detect archaic introgression, achieving robust performance across diverse demographic scenarios.
Oct 2022 - Sep 2023	Undergraduate Researcher. ADVISED BY LUIS A. GARCIA University of Southern California, Department of Computer Science • Evaluated generalized stress detection for health workers using wearable devices across three datasets with shared stress representations. • Demonstrated gradient boosting trees outperform linear support vector machines, random forest, and feed-forward neural networks for continuous stress detection.

Jun 2020 -
Jan 2021

Undergraduate Researcher.

ADVISED BY ELIA TAIT-WOJNO

University of Washington, Department of Immunology

- Elucidated how the PGD2-CRTH2 pathway suppresses Type 2 intestinal immune response during helminth infections in murine models with single-cell RNA sequencing analysis.
- Identified canonical markers of epithelial immune cell types and characterized gene expression in inflammatory immune response in murine cecum during infection.

Jul 2019 -
Aug 2019

Research Intern.

ADVISED BY JAKE VALENZUELA

The Institute for Systems Biology, Baliga Lab

- Developed electroporation protocol that facilitates the transfer of CRISPR-Cas9 complex into *C. reinhardtii*.
- Created and presented how nitrogen starvation increases lipid production in *C. reinhardtii*.

Work Experience

Apr 2025 -
Present

Program Lead

STANFORD SUMMER RESEARCH PROGRAM

- Led program development and coordination for undergraduate research interns from underrepresented institutions, with a focus on graduate school preparation.
- Designed and taught an introductory programming workshop for wet lab researchers, introducing core data science concepts.

Jun 2021 -
July 2023

Viterbi Student Ambassador, Content Lead.

USC VITERBI SCHOOL OF ENGINEERING ADMISSIONS

- Led team of 12 students to produce bi-weekly virtual student panels about student life for audiences of 200+ prospective engineering students.
- Advertised panels through social media outreach, bi-weekly YouTube videos, and Spotify podcasts.

Aug 2022 -
May 2023

Artificial Intelligence Intern.

THE ELLISON INSTITUTE FOR TRANSFORMATIVE MEDICINE

- Developed deep learning models to automate the diagnosis for breast and prostate cancer from digital pathology.
- Refined quality control pipeline to process thousands of whole slide images (WSI) in the cloud.

May 2022 -
Aug 2022

Software Engineering Intern.

ORACLE CLOUD INFRASTRUCTURE

- Designed and tested automated daily health checks for cloud billing accounts.

Awards & Honors

USC Discovery Scholar 2024
USC Google Alumni Award in Quantitative Biology 2024
Knight-Hennessy Scholarship Finalist 2024
Barry Goldwater Scholarship 2023
Astronaut Scholarship Finalist 2023
USC Viterbi Dean's List 2020-2024
USC Viterbi Undergraduate Merit Research Fellowship 2020
USC Presidential Scholarship 2020
USC Dornsife Thematic Option, Reading & Writing 2020

Publications

- [5] Fan C, **Cahoon JL**, Dinh BL, Vecchyo DO, Huber C, Edge MD, Mancuso N, Chiang CWK. "*A likelihood-based framework for demographic inference from genealogical trees.*" *Nature genetics*. 2025 Mar 20.
doi: 10.1038/s41588-025-02129-x.
- [4] **Cahoon JL**, Rui X, Tang E, Simons C, Langie J, Chen M, Lo YC, Chiang CWK. "*Imputation Accuracy Across Global Human Populations.*" *American Journal of Human Genetics*. 2024 May 2. [International Genetic Epidemiology Society 32nd Annual Meeting Best Poster]
doi: 10.1016/j.ajhg.2024.03.011.
- [3] **Cahoon JL**, Garcia L. "*Continuous Stress Monitoring for Healthcare Workers: Evaluating Generalizability Across Real-World Datasets.*" The 14th ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM-BCB). 2023 Sept 3.
doi: 10.1145/3584371.36129741.
- [2] Sheng X, Xia L, **Cahoon JL**, Conti DV, Haiman CA, Kachuri L, Chiang CWK. "*Inverted genomic regions between reference genome builds in humans impact imputation accuracy and decrease the power of association testing.*" *Human Genetics and Genomics Advances*. 2022 Nov 11. [HGGAdvances Award for Outstanding Early Career Publication]
doi: 10.1016/j.xhgg.2022.100159.
- [1] Oyesola OO, et. al. "*PGD2 and CRTH2 counteract Type 2 cytokine-elicited intestinal epithelial responses during helminth infection.*" *Journal of Experimental Medicine*. 2021 Sep 6;218(9):e20202178.
doi: 10.1084/jem.20202178.

Oral Presentations & Other Projects

- May 2025 Self-Guided Query Expansion for Longitudinal Clinical Retrieval, *Biomedical Data Science Student Seminars*, Stanford University.
- May 2025 Impact of Sanctuary Policies on Healthcare Utilization Among U.S. Agricultural Workers: A Causal Inference Approach, *Outcomes Analysis Final Presentation*, Stanford University.
- May 2023 Imputation Efficacy Across Global Human Populations, *Southern California Evolutionary Genetics and Genomics Meeting 2023*, University of California Irvine.
- Dec 2022 Assessing Imputation Quality for Diverse Populations, *Department Research Seminar, Center for Genetic Epidemiology* Keck School of Medicine.
- Nov 2022 Predicting Foster Care Outcomes in the United States with the National Youth in Transition Database, *Artificial Intelligence for Sustainable Development Final Presentation*, University of Southern California.

Teaching

- Jun - Aug 2025 Mentored two undergraduate interns in research communication and graduate school advising
- Jan - Apr 2023 Project mentor for undergraduate machine learning projects through CAIS++
- Sep - Dec 2022 Curriculum Lead for open source deep learning curriculum for undergraduates
- Jan - May 2022 Course Producer for CSCI 104, Data Structures and Objected Oriented Programming

Leadership

- Aug 2022 - May 2023 **Co-President.** The Center for Artificial Intelligence's Student Branch (CAIS++)
Directed all organization initiatives, including 5-8 semester projects, 4 Fall curriculum groups, two project showcases, weekly general meetings, and speaker events to engage undergraduates in artificial intelligence.
- Sep 2021 - May 2022 **Project Manager.** Novus Think Tank. Oversaw six focus project groups that targeted critical social issues impacting the university and surrounding areas.